Suleyman Musayev

Fraud & Abuse Engineer · Detection, Account Security & PKI

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SUMMARY

Fraud and abuse engineer specializing in detection and account security for high-risk environments. Sole architect of the authentication and anti-fraud platform at SSL.com (top-10 global certificate authority): bot defense at 1,000+ fake signups/day, insider-threat and identity-theft detection, and long-horizon surveillance of attackers probing authorization and account flows. Stopped a four-figure-per-day SMS pumping attack in 4 days; defeated a persistent session hijacking attacker. I pair LLM-augmented pattern matching with human-led forensics to take an attack from detection to mitigation in minutes. Open to senior and staff fraud, abuse, trust & safety, and security engineering roles.

EXPERIENCE

Senior Staff Engineer · SSL.com

Aug 2023 - Present

Remote · top-10 global PKI vendor · sole architect of auth & identity platform

- **Detected and shut down months-long attacker reconnaissance** across authorization and account flows, closing every probed vector in code before exploitation - including an account-takeover path in account recovery, a second-factor bypass in the authentication flow, and abuse of the invitation system used to push offensive content.
- Built **bot-account defense handling 1,000+ fake signups per day**, including countermeasures against human CAPTCHA-solving farms; deployed reCAPTCHA v2/v3 and Cloudflare Turnstile.
- Built **insider-threat detection** covering privileged-account misuse and impersonation of staff and admin roles.
- Built **identity-theft detection** that cross-matches IP geolocation, billing, shipping, personal details, and phone numbers across multiple applications.
- **Stopped a four-figure-per-day SMS pumping attack**: designed, built, and deployed the countermeasure in 4 days using dynamic risk scoring, anomaly detection, and adaptive throttling.
- **Defeated a persistent session hijacking attacker** who survived conventional account-recovery and endpoint remediation: built session binding, replay attack detection, and automatic termination of suspect sessions at the application layer.
- **Sole architect** of the authentication and identity platform supporting tens of thousands of daily users; scaled the core business 10× across TLS, code signing, and identity certificate issuance.
- Designed and implemented **OAuth 2.0, OpenID Connect (OIDC), SAML, and session-based** auth, including secure cross-domain flows and external identity provider integrations.
- Built MPIC (Multi-Perspective Issuance Corroboration) compliance per CA/Browser Forum Baseline Requirements; led architecture for distributed auth services, API servers, and audit-logging pipelines.

Software Engineer · Midstay

Aug 2022 - Jan 2024

Remote · early engineer, 5-person team, ~1k DAU consumer product

- Closed **authorization flaws surfaced in a live penetration test**, including a broken access control in a private messaging flow.
- Led incident response against exploitation attempts by advanced attackers; implemented safeguards to prevent recurrence and improve system resilience.
- Reduced backend latency by up to **5×** and improved user retention by **15%** through performance and product improvements.
- Shipped 4 major product features in 12 months, supporting multiple product pivots.
- Final 5 months were a contract wind-down with gradual transition of responsibilities to the rest of the team while ramping up at SSL.com.

Quantitative Trading Operations · Private Trading Firm

Mar 2020 - May 2021

Risk management for leveraged portfolio across brokers, markets, and instruments

- Built automated hedging strategies; reduced hedging costs by up to **35%** through strategy optimization and execution improvements.
- Developed strong intuition for risk modeling and decision-making in high-stakes environments - a foundation I apply directly to security threat modeling.

SKILLS

FRAUD & ABUSE	Fraud Detection, Bot Mitigation, Insider Threat Detection, Identity Theft Detection, Account Takeover Defense, Session Security, Replay Attack Mitigation, Device Fingerprinting, Behavioral Analytics, IP Intelligence, Velocity Controls, reCAPTCHA v2/v3, Cloudflare Turnstile
AUTH & IDENTITY	OAuth 2.0, OpenID Connect (OIDC), SAML, JWT, Session-based, SSO, mTLS, Cross-domain Auth, External IdP integration
SECURITY & PKI	PKI, Certificate Lifecycle Management, MPIC, CVE-aware Defense
CRYPTOGRAPHY	Memory-hard hashing (Balloon, Yescrypt), Certificate-based Encryption, X.509, TLS
LANGUAGES	Ruby, JavaScript, Python, SQL, C, C++
FRAMEWORKS	Ruby on Rails (4-8), Sinatra, Stimulus, Hotwire
DATA	PostgreSQL, Redis, Sidekiq
INFRASTRUCTURE	Docker, Linux, Nginx, AWS, GitHub Actions, CI/CD

OPEN SOURCE

balloon_hashing & yescrypt

Ruby implementations of the Balloon and Yescrypt memory-hard password hashing algorithms - modern, GPU-resistant alternatives to bcrypt.

omniauth_oidc - github.com/msuliq/omniauth_oidc

OmniAuth strategy for OpenID Connect with first-class support for Microsoft Entra ID and Cloudflare Access.

ip_geo_lookup - github.com/msuliq/ip_geo_lookup

IP geolocation lookup for Ruby designed for fraud detection, access control, and request enrichment in security workflows.

Contributions

Authlogic (Ruby authentication framework), openmpic (Multi-Perspective Issuance Corroboration reference implementation).

WRITING & LANGUAGES

"How to Stop SMS Pump Fraud From Draining Your Wallet (and Sanity)" - Medium.

Languages: English (Professional) · Russian (Native) · Turkmen (Native) · Turkish (Conversational)